Cutting S1-B — Visual Assembly Manual

Revision AM-PRE-1 | build 2026-10-29 | PRE-RELEASE BUILD/QUALIFICATION MANUAL

This is the single canonical assembly manual for the S1-B wire cutting/strip machine. It is a controlled pre-release document: the design is defined in the released specifications, but physical fabrication and qualification are NOT complete. Nothing in this manual is presented as a completed qualification record. Every section carries its status; open gates are listed in section 10 and must close on their named physical tests before the machine is used for production.

1. Document control

field	value
Document	release/Cutting_Assembly_Manual.pdf (the one canonical PDF)
Source commit	branch completion/2026-10-29 (pinned base 0f28aa3)
BOM authority	bom/*.csv (sha256/16: selected-cutting-head 910edfd8af8b012b)
Spec authority	shared/*.hpp (sha256/16: controls_spec 0fbe8fdb1fbb46d6)
Signal contract	requirements/signal-table.md (sha256/16: e160b31ff24911bd)
Software	software/machine (host tests green incl. observability/waste/negative-control groups)
Status vocabulary	DEFINED / NOT FABRICATED / OPEN GATE (see handoff 14)
Not included	no qualification results, no delivered-cost claim, no safety certification

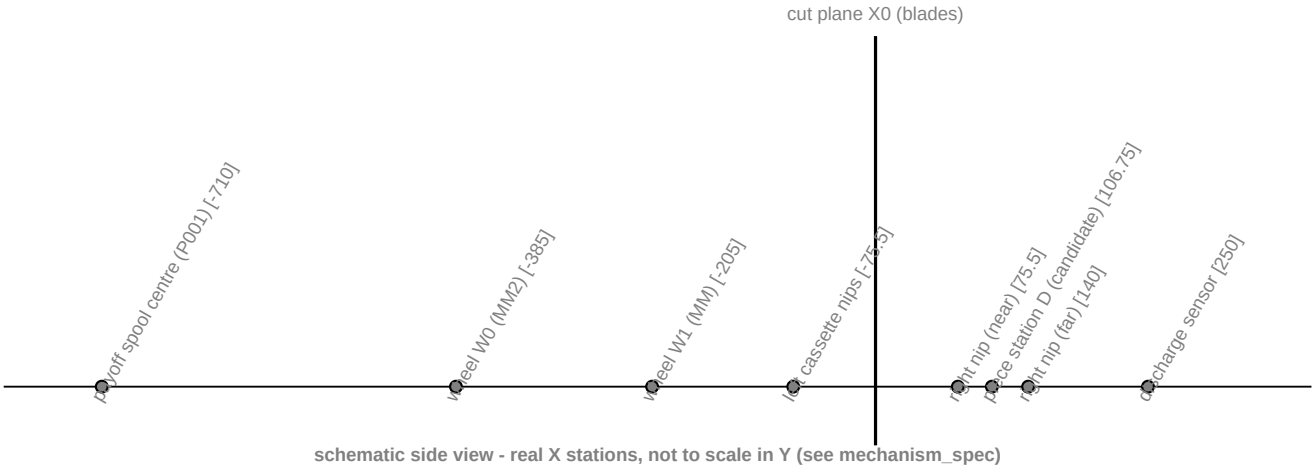
2. Safety warnings (read before any construction work)

WARNING — the items below are construction and operating hazards. They are not a substitute for the risk assessment and safety validation (P03/P04/P12) that remain open.

- Mains and the 36/60 V DC buses carry lethal energy. Lock out and verify de-energised before opening the cabinet.
- The cutting blades are hardened, sharp and spring/cam driven. Never reach into the head; the E-stop removes energy but stored energy must be released (M21/M37 records).
- The payoff spool weighs up to 30 kg (plus wire). Two-person lift or a lifting aid; the spool retention and brake are NOT safety brakes and provide no guaranteed stop (docs/26).
- The blade/axis proof loads (2500 N axial class) are structural loads for the machine, never body loads. No person may occupy the load path during any powered test.
- Waste stubs/slugs are ejected into a sweep chute. The chute sensor is a handling channel, not a safety device.
- Electrical diagrams in this manual are functional references; the electrical release (branch protection, conductor sizing, terminal torque) is an OPEN GATE (G32).

3. Machine overview and station map

Coordinate contract: X = wire travel, Y = opposing cutting strokes, Z = up; cut plane X0; wire at Y0/Z112; +X toward the piece/eject side. All dimensions in mm. The view below shows every real X station used by the sensing/handling contract.



schematic side view - real X stations, not to scale in Y (see mechanism_spec)

Stations (real coordinates): payoff spool centre X=-710; measuring wheels W0/W1 at X=-385/-205 (stock side - they can never observe a severed piece); left cassette nips at X=-75.5/-140; right cassette nips at X=75.5/140 (a severed body shorter than 140 mm cannot be propelled by them); candidate piece station D at X=106.75; discharge clearance sensor zone X=230..270; waste chute sensor is on the fall path below the blade pocket.

4. Subassembly work structure

Build order: (A) base and frame; (B) two cutting axes; (C) head and blade tooling; (D) feed cassettes and measuring modules; (E) payoff; (F) guards; (G) cabinet and wiring; (H) commissioning. Each section below gives identities, fasteners, tools, inspection points and warnings. Torque values shown as PENDING are open gates: do not guess them.

4.1 (A) Base and frame — H001/H002 + welded frame

item	identity	qty	fasteners / torque	inspection	status
base datum	H001 ground datum 180x1040x20 steel	1	M12 floor anchors PENDING (site)	flatness to drawing PENDING	DEFINED / NOT FABRICATED
risers	H002 riser 110x355x20 steel	2	per drawing PENDING	stage grid 46/100 verify	DEFINED / NOT FABRICATED
frame	welded frame (mechanism frame group FR_)	1	weld procedure PENDING	weld inspection PENDING	DEFINED / NOT FABRICATED

4.2 (B) Cutting axes — AX-01 x2 (see exploded schematic)

balloon	identity	qty/axis	fasteners / torque	inspection	status
1	AX-R01 HGR20 400 rail + 2x HGH20CA	2	M5 rail bolts at 60 pitch; torque PENDING	rail straightness; block preload	DEFINED / NOT FABRICA
2	H006 carriage plate 60x75 pad at 46	1	4x HGH20CA lands; ISO4762 M5x16 cl.10.9 (HF0-B) [illegible]	[illegible]	DEFINED / NOT FABRICA
3	AX-S01 SFU1610 C7 500, 10 mm lead + DSG16H	1	nut flange screws PENDING	backlash/runout PENDING; end machining	DEFINED / NOT FABRICA
4	BK12/BF12 supports	1 set	M5 lands; torque PENDING	support alignment PENDING	DEFINED / NOT FABRICA
5	DR-01 34E1K-85 + CL86T-V41	1	HM-B ISO4762 M5x25 cl.10.9 + HM-W [illegible]	[illegible]	[illegible]
6	CP-01 R+W MK2/100/50/8/14 (10 m)	1	hub clamping screws per R+W instruction [illegible]	[illegible]	DEFINED / NOT FABRICA
7	H005 base plate 20x86x440	1	6x 6.6 base grid 46/100; torque PENDING	rail seats milled; BK/BF lands	DEFINED / NOT FABRICA

WARNING — proof loading (2500 N axial / 750 N transverse / 375 N vertical, double) is a controlled validation procedure (P05), not an assembly step. The 4900 N class support rating is below the 5000 N proof load: this is an OPEN GATE (G37) and must never be closed by rounding.

4.3 (C) Head and blade tooling — H004/T001/H007

item	identity	qty	fasteners / torque	inspection	status
holder	H004 monolithic holder 42CrMo4+QT 650	2	MPa min yield	per drawing PENDING	DEFINED / NOT FABRICA
blades	T001 matched 90 deg V pair, 49x38x3	1 pair	shimmed seat; clamp torque PENDING	edge radius; loaded face gap (G32)	DEFINED / NOT FABRICA
backing	H007 backing plate 30x46x8	2	torque PENDING	shim record; 1200 N blade reaction	DEFINED / NOT FABRICA

4.4 (D) Feed cassettes and measuring modules

One left (stock) and one right (piece) cassette; two measuring wheel modules W0/W1 on the stock side (X=-385/-205). The downstream piece station D (X=106.75 candidate) is an OPEN GATE (M02 fit study).

4.5 (E) Payoff (docs/26 R2)

Assemble P003 base -> P004/P005 posts with SKF 6204-2Z/C3 bearings -> P002 spindle with M20x1.5 retention -> P008 keyed drive boss welded to the left flange INSIDE the hub tube -> P001 spool -> P006/P007 brake. The brake is effective only with P008 fitted. The spool centre sits at X=-710.

5. Wiring and signals

Functional reference only — the electrical release is an OPEN GATE. The authoritative signal contract is requirements/signal-table.md; summary:

signal	meaning	status
GP0-7	step/dir to 4 drivers via TPL7407LDR sinks	DEFINED
GP8-11	W0/W1 quadrature via AM26LV32E (ifm RU3100, TTL 2048 PPR)	DEFINED
GP12-16,18-21,27	panel/safety/ALM/home coupler inputs (LOW = active)	DEFINED
GP17,22,26,28	drive enable, LEDs, buzzer	DEFINED
piece station D	quadrature of the downstream module	OPEN GATE (Phase H pin)
part_clear	discharge present->clear evidence; fails closed while unassigned	OPEN GATE (Phase H pin)
waste_clear / waste_sweep	waste-chute transit + sweep actuator; fails closed	OPEN GATE (Phase H pin)

6. Commissioning and qualification flow

- Flash the firmware image, record its hash, then run the factory-test UF2 (separate image) to exercise every I/O channel.
- Enter service fixture mode only for engineering runs (never emits accepted counts or approval).
- Home both blades; route wire through the cassettes; run the commissioning polarity moves; calibrate all three measuring stations (W0/W1/D).
- Approve the retained recipe over the host link only after commissioning; production START requires this approval.
- Execute the physical qualification: P00-P15 families plus the 18 TS and 17 TF thermal tests (tests/P10-thermal/procedures.md).
- Close every open gate in section 10 with signed records; only then may this manual's status change.

7. Maintenance (provisional — P14/TS evidence owed)

item	interval	limit / action	status
filters (two-path ventilation)	inspect each shift; replace per TS evidence	flow-service limit triggers change	OPEN GATE
blade edges T001	inspect per shift	edge radius/wear limit from coupons (G03)	OPEN GATE
measuring wheels W0/W1/D	weekly	tread wear/calibration drift check (G24)	OPEN GATE
nips and guides	weekly	50-75 N nip force check (G23)	OPEN GATE
payoff brake pads	monthly	friction pad wear (docs/26)	OPEN GATE
bearing lubrication	per P14 duty test	interval to be established	OPEN GATE

8. Fault and recovery (operator actions)

fault	meaning	operator action
red LED steady	latched fault; drives disabled	clear cause; hold RESET 0.5 s; rehome
no_wire / length_slip	material motion did not match the plan	inspect wire path and wheels; rethread; RESET
part_not_cleared	discharge evidence missing (quarantine)	remove/verify the part in the takeaway; log the scrap
waste_not_cleared	stub/slug did not transit the chute	clear the sweep chute and pocket; RESET
wire_geometry	planned move/score rejected by the wire model	do not force; rethread and rehome; report
power loss mid-piece	in-flight part has uncertain state	on restart: rehome + rethread; the prime pass scrap-discharges leftovers

9. Spare parts (initial list)

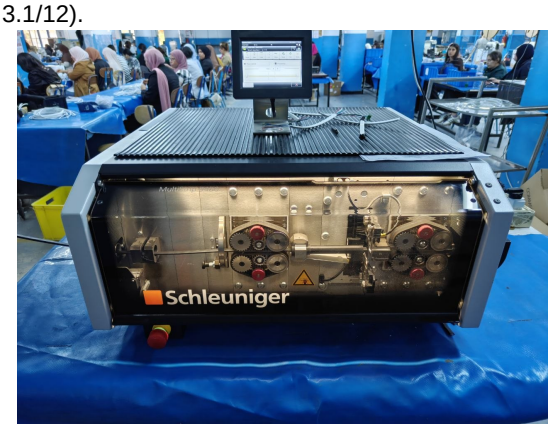
part	identity	qty	status
blades	T001 matched 90 deg V pair	2 pairs	DEFINED / NOT FABRICATED
filters	Pfannenberg 11666102055 / 11760002055 candidates	1 set	OPEN GATE (TS evidence)
measuring wheels	ifm E60137	1	DEFINED
encoders	ifm RU3100	1	DEFINED
fuses/relays	per electrical release	-	OPEN GATE (G32)

10. Open-gate register (must close before production)

gate	what is owed	record
G37 axis loads	revalidate acceptance loads vs the frozen 10 mm-lead screw; resolve 4900 N < 5000 N proof	P05
G16 fits	3-6 um interference matched pairs + bearing clearance envelope	P01/P13
G18 shaft thermal	shaft rise <= 20 K above housing (stored 29.66 K case fails)	P10
G22 face gap	0.020 mm loaded gap interval from cut-quality data	P07/P13
G29-G30 ventilation	installed filtered flow and thermal margins; 18 TS + 17 TF	P10 + tests/P10-thermal
G32 protection	exact fuse/breaker coordination and wire/terminal limits	P02/P04
G35 safety	risk-derived PLr/SIL, dual-channel validation by a competent safety engineer	P03/P04/P12
Phase H I/O	assign part_clear / waste_clear / waste_sweep / piece-station pins	signal-table
G39 RFQs	send the drafted enquiries (couplings, axis, protection, custom work, Tunisia delivery)	procurement records
G47 cost	complete landed cost + 70 % like-for-like comparison	cost rollup

11. Photographic reference

Layout-inspiration photos from the machine under construction. They are reference images only: they establish no ratings, dimensions or qualification (handoff 3.1/12).



front view (reference)



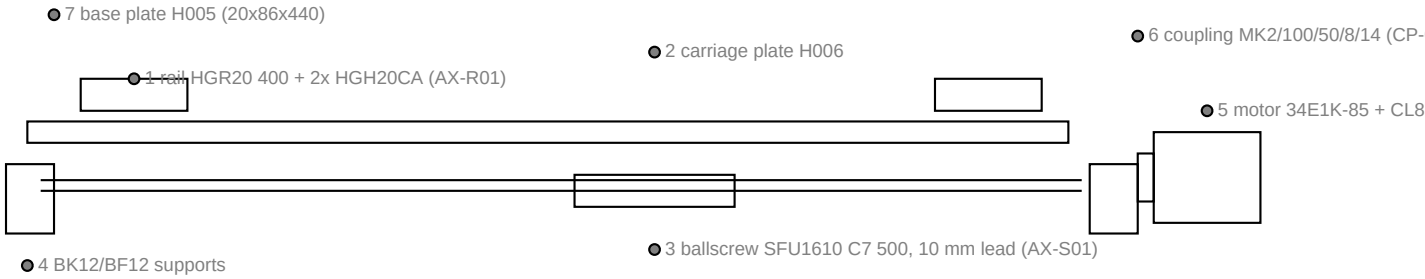
rear view (reference)



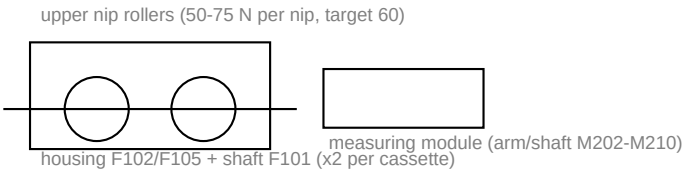
payoff and buffer (reference)

12. Vector views

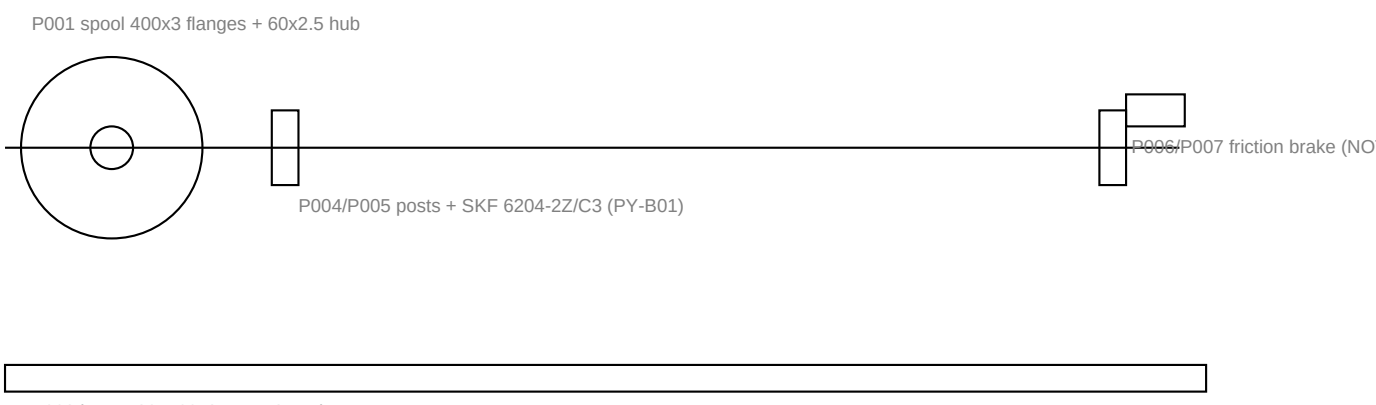
Schematic vector views generated from the released spec geometry with real station coordinates and ballooned BOM identities. They are assembly aids, not native-CAD renders; the native CAD views remain part of the release gate and must not be substituted by these schematics.



exploded schematic of AX-01 (one of two axes) - ballooned to the BOM



feed cassette schematic (F-series) - forces per G23



payoff schematic (docs/26 R2) - <=30 kg spool; keyed boss P008 torque path